Numerical Verification of a Curvature-Bound Bounce Cosmology: Effective-Field-Theory Control Pushes the Transition Below the Planck Scale

Ho Hyung Kim

Independent Researcher, Seoul, Republic of Korea

iconarea@gmail.com

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Abstract

Classical general relativity can lead to singularities of divergent curvature and density in black-hole interiors and in a backward extrapolation of the early Universe. This paper takes a minimal phenomenological framework — a coordinate-invariant curvature bound, an effective non-singular bounce at a critical density, and a distinction between an event horizon and the genuinely high-curvature region — and reports what happens when its entries are computed rather than asserted. The equation-of-state transition is among them, at the level of a fluid: constant-w matter and a scalar oscillating about a minimum are both excluded, and what survives — energy conversion above a density threshold — yields a lower bound on the conversion rate, the prediction $p_f = 3/2$, and a new requirement $\rho_H/\rho_c > (\rho_i/\rho_c)^{5/13}$. No covariant action generating that transition is available: a conformal coupling and the beyond-Horndeski reconstruction both close, for different reasons, and only a coupling to the kinetic term survives as a candidate whose background equations have not yet been obtained by variation, so those conclusions remain conditional on a fluid description. So is the location of the curvature bound: it is not assumed to be Planckian but computed, and effective-field-theory control places it six to sixteen orders of magnitude below the Planck curvature, at $\rho_c^{1/4} \sim 10^{16}$–$10^{17}$ GeV. The effective bounce background is shown to coincide *exactly* with a background ansatz for which a stable $c_T = 1$ beyond-Horndeski construction already exists in the literature, with an effective Friedmann residual of 2.8×10^{-17}. Within that construction a $w(t)$ transition background is ghost-free, gradient-stable and subluminal throughout, and suppresses shear. Two routes are excluded in the process: the minimal single-field realisation, and the constant-equation-of-state baseline, the latter because scalar subluminality requires $w \leq 1$ while shear suppression requires $w > 1$. Integrating the Mukhanov–Sasaki equation with the z''/z that follows from the surviving trajectory gives a strongly blue tilt, so the adiabatic mode of the bounce cannot originate the observed fluctuations — and on the derived background it moves further away still, from $n_s = 3.34$ to 4.54; an entropic mechanism repairs this, on the derived background too, at the unchanged cost of tuning a tachyonic entropy mass to 0.29% — though the tuned value itself must be re-solved per background. The coherent-relic envelope is shown *not* to be derivable from the background: the transfer function carries no band-limited feature, and an isotropic background cannot generate angular structure at all. The only dynamical source of a direction is residual shear, which forces a quadrupole with a predicted scale dependence $g_*(k) \propto k^{-0.70}$, reducing the observational template from six free parameters to two. The amplitude remains unpredicted, so only an upper limit follows. No observational data was used in this work.

Keywords: quantum gravity; Planck scale; curvature invariant; bounce cosmology; beyond-Horndeski; statistical anisotropy; early Universe

1 Problem statement

In general relativity, gravity is spacetime curvature rather than a single force strength. The classical interior solution of a black hole and a backward extrapolation of FLRW cosmology both reach, under stated conditions, a singularity of divergent curvature. It is more conservative to read that divergence as a signal that the classical theory is insufficient in the region than as an actual infinity in nature.

One distinction matters throughout. The **event horizon** of a large black hole can be locally weakly curved and is not automatically a quantum-gravity region. The transition candidate considered here is not the horizon but the region where spacetime curvature or matter density approaches the Planck scale.

2 Assumptions and scales

2.1 Planck-scale benchmark

Combining $\hbar$, G and c gives the characteristic length and density

$$\ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}}, \qquad \rho_{\mathrm{P}} = \frac{c^5}{\hbar G^2}. \tag{1}$$

These are dimensional benchmarks at which quantum effects and gravity are expected to become simultaneously non-negligible, not a proof of a unified theory.

2.2 Curvature-bound hypothesis

A "maximum gravity" is properly expressed through a coordinate-invariant curvature scalar rather than an observer-dependent acceleration. For the Kretschmann scalar $\mathcal{K} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ we assume that a *finite* bound exists,

$$\mathcal{K} \leq \mathcal{K}_{\max}. \tag{2}$$

This is the only curvature assumption made, and no microscopic origin for it is supplied.

The location of the bound is not assumed. The Planck length is a natural unit in which to express equation (2), not a claim that the bound sits there. Defining

$$\eta \equiv \mathcal{K}_{\max}\, \ell_{\mathrm{P}}^4 \tag{3}$$

makes η a quantity to be computed rather than declared. Section 7 shows that requiring effective-field-theory control gives $\eta \sim 10^{-16}$–10^{-6}, so the transition occurs six to sixteen orders of magnitude below the Planck curvature, at $\rho_c^{1/4} \sim 10^{16}$–$10^{17}$ GeV.

The proposition this section leaves is therefore not that a bound exists at the Planck scale, but that *a finite bound exists and its location is fixed by the self-consistency of the effective description* — a narrower and more testable statement.

2.3 String length scale and the high-curvature crossover

In perturbative string theory the fundamental length is $\ell_s = \sqrt{\alpha'}$. The low-energy derivative expansion is trustworthy for $\alpha'|R_{\mu\nu\rho\sigma}| \ll 1$; when $\alpha'|R_{\mu\nu\rho\sigma}| \gtrsim 1$, higher-curvature corrections and stringy states cannot be neglected. This motivates, but does not establish, the weaker hypothesis used here: that new degrees of freedom near the string scale may produce a finite effective curvature or a non-singular transition.

For observability, causality is imposed on the coherent interval rather than on the Universe as a whole,

$$L_{\mathrm{coh}}(t) \leq D_{\mathrm{causal}}(t), \tag{4}$$

where D_{causal} is the particle or event horizon of the chosen cosmology.

2.4 Causality and unitarity

The framework does not assume that causality is lost beyond an event horizon. It assumes instead that a more fundamental quantum causal rule or unitary evolution governs the transition region where the classical description fails. The explicit form of that rule is the central gap of the proposal.

3 An effective bounce ansatz

For a spatially flat homogeneous universe with $c = 1$, the simplest phenomenological effective equation is

$$H^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_c}\right), \qquad \rho_c = \xi\rho_{\mathrm{P}}, \tag{5}$$

with ξ a dimensionless coefficient that a microscopic theory must supply. Combined with $\dot{\rho} + 3H(\rho + p) = 0$ this gives

$$\dot{H} = -4\pi G(\rho + p)\left(1 - 2\frac{\rho}{\rho_c}\right), \tag{6}$$

so that for ordinary matter with $\rho + p > 0$ one has $\dot{H} > 0$ at $\rho = \rho_c$, permitting a non-singular transition from $H < 0$ to $H > 0$.

Equation (5) is an *illustrative ansatz* resembling effective forms widely studied in bouncing cosmology. Writing it down does not derive loop quantum gravity, string theory, or any other specific proposal. Section 7 reports what happens when one asks which microscopic realisations are actually compatible with it.

4 Black holes, the Big Bang, and a new expanding region

The minimal narrative of the hypothesis is: gravitational collapse or a black-hole interior evolves to high density; curvature and density approach the Planck bound; a quantum-gravity transition replaces the classical singularity; a bounce occurs; an expanding spacetime region follows.

If that final region is causally disconnected from the parent spacetime, an observer inside it could see a hot dense early universe resembling a Big Bang. This is *not* the claim that the external Hawking evaporation of one black hole is observed as our Big Bang; the externally visible evaporation and the hypothesised interior bounce are distinct problems. Nor does the framework claim that a black hole promptly becomes a white hole in the same external universe: equation (5) is a homogeneous ansatz and cannot simply be inserted into a black-hole interior. A separate spherically symmetric or rotating solution, a transition timescale, boundary conditions on energy and information, and an explicit causal structure would all be required.

Mass should likewise be described as a conversion between hot field and radiation energy and particle masses, not as creation from nothing. Which species are produced, and whether any stable electromagnetically inert remnant could constitute dark matter, requires a separate quantum-field-theoretic calculation.

5 Relation to other proposals

Conformal cyclic cosmology connects the far future of one aeon to the Big Bang of the next by a conformal rescaling. The model here does not assume a conformal matching; it focuses on a dynamical rebound at maximum curvature or critical density.

Ekpyrotic and cyclic models are closer, in that they treat a pre-Big-Bang contracting phase and a transition. This work does not assume brane collisions or a specific higher-dimensional theory. At this stage it is therefore not a replacement for those models but a statement of the singularity-avoidance and transition rules they all require.

6 Falsifiability and observational program

To become a scientific model, the framework must make at least one quantitative prediction differing from standard inflationary cosmology.

1. **Primordial CMB fluctuations.** *Partly supplied* (§7). The adiabatic tilt ($n_s = 3.34$ on the earlier background, 4.54 on the derived one) is an exclusion; the quadrupolar modulation $g_*(k) \propto k^{-0.70}$ is a signal whose *shape* is predicted. Its amplitude is not.

2. **Primordial gravitational waves.** *Not supplied.* The surviving construction has $c_T^2 = 1$ by design and is therefore compatible with GW170817, but that is a constraint the construction was built to satisfy, not a prediction.

3. **Nucleosynthesis and structure formation.** A reheating history compatible with light-element abundances. Not addressed here.

4. **Black-hole observations.** A mass remnant, ringdown deviation, or echo signature cannot be claimed without an explicit matching rule, which §4 does not supply.

6.1 Indirect JWST route

JWST cannot observe Planck-scale curvature directly. It can constrain early black-hole seeds and their hosts — for instance through a uniform census of $z \gtrsim 4$ Little Red Dot candidates in public NIRCam and NIRSpec data, jointly fitting spectroscopic redshift, line width, host stellar mass and lensing selection. Such a program compares astrophysical seed channels; by itself it cannot establish a bounce, a wormhole, or a curvature bound.

7 Numerical verification

This section reports only those items of §6 that were actually computed. No observational data was used.

7.1 The background lies inside a known stable family

In units $\alpha = 1$, $a_B = 1$, the exact constant-w solution of equation (5) is

$$H(t) = \frac{t}{p\,(1+t^2)}, \qquad a(t) = (1+t^2)^{1/(2p)}, \qquad p = \tfrac{3}{2}(1+w), \tag{7}$$

which is *exactly* the background ansatz of Ye and Piao [1] at constant p and unit lapse — the family for which they give an explicit $c_T = 1$ beyond-Horndeski construction. The residual of equation (5) on this solution is 2.8×10^{-17}: an identity, not a fit. The ansatz of §3 is therefore not arbitrary phenomenology.

7.2 Stability

Two candidates are excluded. A minimally coupled single field $P(X) = KX + LX^2 - V$ yields no point, out of 10^6 sampled, that violates the null energy condition while remaining ghost-free and gradient-stable — consistent with the Horndeski no-go theorem [2]. The constant-w baseline fails for a subtler reason: scalar subluminality requires

$$c_s^2(+\infty) = p/3 \le 1 \iff w \le 1, \tag{8}$$

whereas suppressing anisotropy through contraction requires

$$\Delta \ln(\sigma^2/\rho) = (3/p - 1)\ln(1+\eta_i^2) < 0 \iff w > 1. \tag{9}$$

Condition	Value	Verdict
No ghost (scalar)	$\min Q_s = 3.000$	pass
No ghost (tensor)	$\min Q_T = 0.500$	pass
Gradient stability	$\min c_s^2 = 0.114$	pass
Subluminality	$\max c_s^2 = 0.849$	pass
Tensor speed	$c_T^2 = 1$	by construction
Shear suppression	$\Delta \ln(\sigma^2/\rho) = -6.56$	pass
EFT control	Λ assumed, not derived	**not counted**

Table 1: Stability of the surviving $w(t)$ transition realisation. The final row is the one entry that passes only because a cutoff was supplied by hand; it is therefore not counted as passed.

The two conditions do not overlap. Both boundaries are reproduced to four digits by an exhaustive parameter scan.

A $w(t)$ transition background ($w_i \to 4.33$, $w_f = 0.6$) passes every condition, as summarised in Table 1.

7.3 Primordial spectrum: the adiabatic mode is excluded

Integrating $v_k'' + (c_s^2 k^2 - z''/z)v_k = 0$ with the z''/z that follows from the surviving coefficient trajectory gives $n_s \simeq 3.35$, incomparable with the Planck value $n_s = 0.9649 \pm 0.0042$ [3]. This is a known property of ekpyrotic contraction rather than a surprise. The pipeline was validated against the analytic contracting-phase result

$$n_s - 1 = 3 - 2\left|\beta - \tfrac{1}{2}\right|, \qquad \beta = \frac{1}{p_i - 1}, \tag{10}$$

which it reproduces to within 0.07 over a factor five in p_i.

The adiabatic mode of the bounce therefore cannot originate the observed fluctuations. The standard remedy — an entropic mechanism — was tested on the same background. A tachyonic entropy mass $m_{\text{eff}}^2 = -\lambda H^2$ gives $n_s = 0.9925$ at $\lambda = 104$ and $n_s = 0.9572$ at $\lambda = 106.6$. Since $\mathrm{d}n_s/\mathrm{d}\lambda = -\beta^2/\nu$, matching Planck within its error bar fixes λ to 0.29%. That tuning is the price of the repair, and no mechanism in this framework explains it.

7.4 The coherent relic: shape predicted, amplitude not

A classical relic obeys the same linear mode equation as the fluctuations, so its envelope is the initial profile multiplied by the transfer function, and only the transfer function is background dynamics. The measured transfer function carries *no* band-limited feature: its cutoff lies two orders of magnitude below the comoving scale of the bounce, so every observable mode is far longer than the transition and carries no record of having crossed it. Splitting the transition into two timescales — as a three-level synchronisation hierarchy would imply — changes nothing, because producing a peak requires modes to exit, re-enter and exit again, and $|aH|$ remains monotonic in topology.

The angular structure is a stronger statement. On an isotropic FLRW background the mode equation depends on $|k|$ alone, so no direction dependence can be generated at all. Within this framework the only dynamical source of a direction is residual shear, and it *forces*

$$P(k, \hat{\mathbf{k}}) = P_{\text{iso}}(k)\left[1 + g_*(k)\,(\hat{\mathbf{k}}\cdot\hat{\mathbf{n}})^2\right], \qquad \frac{\mathrm{d}\ln g_*}{\mathrm{d}\ln k} = -0.70. \tag{11}$$

The multipole is fixed at $L = 2$; the modulation is multiplicative rather than additive; and the tilt agrees across two shear amplitudes, so the shape is a prediction. The $\mu = (\hat{\mathbf{k}}\cdot\hat{\mathbf{n}})^2$ dependence was measured rather than assumed, and is linear to within 4%.

The free parameters of the observational template accordingly drop from six to two: an overall size and an axis. The initial shear amplitude is *not* fixed by the theory, so only an upper limit follows and detectability is not predicted. Using the Planck quadrupole constraint $\Delta g_* = 0.016$ [4], the shear amplitude obeys $C \lesssim 2.3 \times 10^{-6}$. That published limit assumes a scale-independent g_*, so this number is indicative pending a dedicated reanalysis with the predicted tilt.

7.5 What a black-hole-interior origin would require

If the reading of §4 is maintained, the level of anisotropy becomes the obstacle. A black-hole interior is Kantowski–Sachs rather than FLRW and starts at $\sigma^2/\rho \sim O(1)$. Reaching the regime verified above ($\sim 10^{-7}$) demands $\Delta \ln(\sigma^2/\rho) \simeq -16$, which at $p_i = 8$ corresponds to a contraction roughly two hundred times longer than the one computed here. The stability verdict would also have to be re-established non-perturbatively on a Bianchi I background. The black-hole–Big-Bang reading is therefore neither excluded nor supported; what has changed is that the requirement is now quantified.

7.6 Joining ekpyrotic and Hagedorn phases, and a route to ξ

After the exclusion of constant w, the open question is what supplies $w(t)$. Joining two string-theoretic ingredients makes the requirements fit.

Flattening and smoothing during contraction operate only for $w > 1$, being a competition between curvature a^{-2}, shear a^{-6} and the background $a^{-3(1+w)}$. An ekpyrotic modulus with a steep negative potential supplies this [6]. A Hagedorn phase, by contrast, puts energy into oscillator and winding modes and has $w \simeq 0$, so it *cannot* flatten — the same point as the known observation that string gas cosmology does not by itself solve the flatness problem [7].

Joined *in sequence*, each phase covers the other's deficit: the ekpyrotic phase flattens, and once the density reaches the Hagedorn scale that phase takes over and supplies the shallow equation of state at the bounce. Tested at $w_f = 0$ (that is, $p_f = 1.5$), the configuration passes every condition of Table 1: $\min Q_s = 3.000$, $\min Q_T = 0.500$, $\min c_s^2 = 0.098$, $\max c_s^2 = 0.820$, $w(-\infty) = 4.33$, $\Delta \ln(\sigma^2/\rho) = -6.55$, with the coefficient gate passed at all 47 999 sampled points.

A possible by-product — a route to deriving ξ — was examined and *closed*. If the Hagedorn scale fixes when the transition occurs, then the EFT cutoff sits at the same place, $\Lambda = M_s = \rho_c^{1/4}$, and is no longer adjustable. Feeding that Λ into the gate gives

$$\frac{E_{\text{char}}}{\Lambda} = 0.519, \tag{12}$$

five times the 0.1 criterion, with 8 833 of 47 999 points violating EFT control. For constant w the analytic value is $E_{\text{char}}/\Lambda = (4/3)^{1/4}/\sqrt{2} = 0.760$, *independent of* w. This is physically unsurprising: if the bounce occurs at the string scale, the characteristic energy of the bounce *is* the string scale, which is not a regime where an effective description is controlled.

Control would require $\rho_H/\rho_c > 725$ — a bounce some seven hundred times below the string density. But then the Hagedorn scale no longer fixes ρ_c, and ξ is free again.

What survives and what does not. **Survives:** the stability of the joined background; ghost and gradient violations remain zero in both sectors even in the run above. **Dies:** the identification $\rho_c = \rho_H$, and with it the route to ξ. **Newly exposed:** if the bounce really is at the string scale, then the stability verdict of §7 was itself obtained with an effective description in a regime where that description is not controlled.

Option (a): push the bounce below the string scale. Accepting $\rho_H/\rho_c > 725$ as a requirement, and returning ξ to being a free parameter, restores control. At $\rho_H/\rho_c = 725$ one finds $E_{\rm char}/\Lambda = 0.1000$ (marginal) and at 10^4, 0.0519 (comfortable), with zero EFT violations; stability is untouched in all cases.

The price is the curvature scale. With $\rho_H/\rho_{\rm P} \sim g_s^4$ one has $\xi_{\rm red} \equiv \rho_c/M_{\rm P}^4 = g_s^4/N$, and since $H = 0$ at the bounce, $\mathcal{K} = 12\dot{H}^2$, so the coefficient η of equation (3) is $\eta = 3\xi_{\rm red}^2(1+w)^2$.

g_s	$N = \rho_H/\rho_c$	$\xi_{\rm red}$	η	$\rho_c^{1/4}$ [GeV]
0.1	725	1.4×10^{-7}	5.7×10^{-14}	4.7×10^{16}
0.3	10^4	8.1×10^{-7}	2.0×10^{-12}	7.3×10^{16}
0.5	725	8.6×10^{-5}	2.2×10^{-8}	2.4×10^{17}
1.0	10^4	1.0×10^{-4}	3.0×10^{-8}	2.4×10^{17}

Table 2: Cost of restoring EFT control by placing the bounce below the string scale. The coefficient η of equation (3) lands far below unity.

Since η falls between 10^{-16} and 10^{-6}, **the curvature bound does not saturate at the Planck curvature** but six to sixteen orders of magnitude below it. The tacit reading of equation (2) with $\eta \sim O(1)$ does not survive; the actual transition scale is $\rho_c^{1/4} \sim 10^{16}$–$10^{17}$ GeV. That is the decade tensor-mode observations probe, though the bounce is not inflation and the bound on r does not transfer directly.

What is gained is EFT control, so the stability verdict is obtained where the description is trustworthy. What is lost is that ξ remains free and the Planck-scale reading of the bound does not hold.

Two caveats remain. (i) $w_f = 0$ approximates the Hagedorn value; the string-gas dynamics was not computed. (ii) The cutoff remains an assumption, and it is now known that the assumption is not harmless. The third caveat of the earlier version — that no equation drives w from 4.33 to 0 — is addressed in Section 7.7.

7.7 The dynamics that drives the transition: two no-gos and one mechanism

The joining above *supplies* $w(t)$ as a tanh profile with four free parameters $(p_i, p_f, \eta_p, \tau_p)$ and reconstructs the rest from it, so "the ekpyrotic phase softens into a Hagedorn phase" was an assumption. Computing it instead gives two exclusions and one surviving mechanism.

Constant-w matter cannot do it. If every component obeys $\rho_i \propto a^{-3(1+w_i)}$, then with weights $f_i = \rho_i/\rho$,

$$\frac{dw_{\rm eff}}{d\ln a} = -3\,{\rm Var}_f(w) \;\leq\; 0 \tag{13}$$

holds exactly. Contraction has $d\ln a < 0$, so $w_{\rm eff}$ can only *increase*, toward $\max_i w_i$. No number of components in any proportion produces the descent $w : 4.33 \to 0$. Equation (13) was verified on a 40,001-point grid to a relative residual of 1.3×10^{-7}.

An oscillating scalar cannot either. The potential $V(\phi) = V_1 e^{-2c\phi} - V_0 e^{-c\phi}$ carries the ekpyrotic tail ($\epsilon = c^2/2$) at large ϕ and turns over into a minimum, about which the virial theorem would give $\langle w \rangle \to 0$. But the oscillator mass and the transition density come from the *same* potential: $m \propto \sqrt{|V_{\rm min}|}$ while the transition occurs at $\rho \sim |V_{\rm min}|$, leaving a time $\propto |V_{\rm min}|^{-1/2}$. The available number of oscillations,

$$N_{\rm osc} = \frac{m\,\Delta t}{2\pi} \;\longrightarrow\; \frac{c}{\pi}\sqrt{\tfrac{2}{3}} = 1.04, \tag{14}$$

is therefore *independent of the depth.* Seven integrations spanning a factor of 100 in depth each gave a single turning point; the end state was kination ($p \simeq 3$), not $p = 3/2$, and deepening the minimum made it worse.

Energy conversion is what remains. Relaxing conservation of the individual components,

$$\dot{\rho}_1 + 3H(1+w_1)\rho_1 = -\Gamma\rho_1, \qquad \dot{\rho}_2 + 3H\rho_2 = +\Gamma\rho_1, \tag{15}$$

makes the descent require

$$\Gamma > 3|H|\, w_1(1-f), \qquad \gamma \equiv \frac{\Gamma}{\sqrt{\rho/3}} \gtrsim 3w_1 = 13. \tag{16}$$

This is the string-gas reading of the Hagedorn transition: the soft phase comes not from a field settling into a minimum but from ekpyrotic energy being converted into string excitations once the density crosses ρ_H. The threshold is on density, not on time, so it carries no width parameter. The numerics confirm the bound — final $w = 2.6 \times 10^{-3}$ at $\gamma = 13$ and 6×10^{-13} at $\gamma = 26$, with the Hamiltonian constraint held to 1.1×10^{-12}.

Gain and cost. Four free parameters become three physical ones (c, ρ_H, γ); $p_f = 3/2$ becomes a *prediction* (measured 1.5000000000009); and the width is fixed by γ alone ($\gamma\tau_p \simeq 44$, unchanged when the initial density varies over six orders of magnitude). The tanh is a 1.1% approximation to the derived shape. The cost is anisotropy: a pressureless stage has $1 - 3/p = -1$ and gives back ekpyrotic suppression by a factor ρ_c/ρ_H, so a net suppression survives only if

$$\frac{\rho_H}{\rho_c} > \left(\frac{\rho_i}{\rho_c}\right)^{5/13}, \tag{17}$$

a bound that did not exist before. Recovering the anisotropy budget -6.55 requires the contraction to begin at $\rho_i \lesssim 2 \times 10^{-10}\rho_c$, ten orders of magnitude below ρ_c rather than the six assumed. Accordingly $p_f = 2.4$ is discarded: the derivation gives 3/2, and keeping 2.4 would need a soft component with $w = 0.6$, which a string gas does not provide.

7.8 Re-running the coefficient gate on the derived background

Since $c_s^2(+\infty) = p/3$ depends on p_f directly, the result above does not leave the earlier stability verdict intact, so the coefficient gate was re-run on the derived background; the unit map between the two calculations checks out, $\max|H|$ agreeing with the closed-form $1/(2p_i) = 1/16$ to eleven decimals. The first run returned $\min c_s^2 = -134.3$, but it was tested for a pole before being recorded and was one: the extrema sat two grid points apart at $H \simeq 0$, because taking the bounce from the sampled maximum of ρ leaves it offset from the true bounce, where the free function $1 - c_2$ and H then vanish at different points and the removable 0/0 becomes an actual pole. After locating $H = 0$ by root-finding, $\mathcal{M}(0)$ matched the analytic $-k_2/4$ to ten decimals. Even so, the earlier coefficients $(k_1, k_2, \tau_1, \tau_2^2) = (12, 5, 1.7, 0.59)$ *fail* on the derived background, $\min c_s^2 = -2.162$ — a genuine gradient instability, not a pole. Re-deriving them recovers a pass: at $(7.5, -6.0, 26.833, 6.75)$, $\min Q_s = 3.0000$, $\min c_s^2 = 0.09258$ and $\max c_s^2 = 0.7838$ over all 480,003 grid points, with 6.19% of 74,529 candidates viable. The substantive change is that k_2 reverses sign. Widening the window by a factor of 25 leaves $\min c_s^2$ unchanged to five decimals. Moving p_f from 2.4 to 3/2 therefore preserves the existence of a stable completion, though not the coefficients that realise it.

7.9 Re-integrating the primordial spectrum

The quoted $n_s \simeq 3.35$ came from the z''/z of the earlier background, so it was re-integrated on the coefficient trajectory above. The first thing encountered was a closed mode window, for a physical reason: the pressureless stage grows the scale factor by $(\rho_c/\rho_H)^{1/3} = 4.64$ between ρ_H and ρ_c, so at a given density $|z''/z|$ is 33 times its earlier value and the smallest wavenumber admitting a WKB vacuum rises above the super-horizon bound at read-out. Pulling the contracting grid ten times further into the past opens a window comparable to the earlier one (0.62 decade against 0.43) — independent evidence pointing the same way as the requirement $\rho_i \lesssim 2 \times 10^{-10}\rho_c$. With the window open, the blue tilt *worsens*: on the same grid the earlier background gives $n_s = 3.343$ and the derived one $n_s = 4.541$. Changing the grid window by a factor of two or ten moves n_s by about 0.05, four per cent of the difference of 1.198, so this is not a grid artefact; the convergence diagnostic is 6.2×10^{-6} over 26 valid modes. Against Planck 2018's $n_s = 0.9649 \pm 0.0042$, $n_s = 4.54$ is further away than 3.34: the adiabatic mode cannot be the origin of the observed fluctuations, and a second field is not optional but required.

7.10 Re-testing the entropic mechanism

Since the adiabatic mode became bluer, the remedy was re-tested here. The mode window narrows for the third time for the same reason — the pressureless stage grows the scale factor by 4.64, raising the WKB bound for the entropy modes too, so the default grid leaves it empty and must be widened eightfold. Scanning λ in $m^2_{\text{eff}} = -\lambda H^2$ over five values there, the mechanism *works*: the measured $n_s(\lambda)$ tracks the analytic $\nu^2 = \frac{1}{4} + \beta(\beta - 1) + \lambda\beta^2$ across the whole range with a nearly constant deviation, an additive shift of the zero point rather than an error in the slope. The *severity* of the tuning is unchanged, 0.293% either way, because the tolerance comes from $\beta = 1/(p_i - 1)$ and $p_i = 8$ is preserved. The *central value* does not carry over: the λ reproducing Planck moves from 106.935 to 108.099, a shift of 3.7 tolerance units, and the offset from the analytic formula grows from -0.0046 to -0.0202 because that formula assumes pure ekpyrotic contraction. Moving p_f to 3/2 therefore leaves the mechanism able to repair the blue spectrum, though the tuned value, like the coefficients, must be re-solved per background.

7.11 Deriving the conversion rate: a failure

The rate Γ was an input above; deriving it from string microphysics closes the route rather than opening it. The two scalings correspond to different mechanisms: $\Gamma \propto \sqrt{\rho}$ is $\Gamma \propto H$, which is not a wrong ansatz but the one appropriate to *gravitational* particle production, where the rate is set by non-adiabaticity expressed through H and $\dot{H}$; a *thermal, collisional* rate is instead set by the local energy scale, $\Gamma = c\rho^{1/4}$. Only the latter is tested here. With the thermal scaling the descent condition becomes density-independent, $C > \sqrt{3}w_1 = 7.506$, sharper than $\gamma > 3w_1 = 13$. A ceiling also appears: the conversion completes only for $\rho_{\text{tr}} \lesssim 0.3\rho_c$, and since the anisotropy budget improves as the threshold rises (from -3.13 to -13.30), 0.3 is best for both and the threshold is squeezed from both sides. And there it closes. EFT control put the bounce below the string density, $\rho_s/\rho_c > 725$, so the transition necessarily sits at 3–14% of the string scale, where the Hagedorn density of states does not cancel the Boltzmann factor — it does so only at $T = T_H$. Producing the heavy states of a pressureless string gas is exponentially suppressed, not enhanced: overcoming it would need an unsuppressed coupling of 59 in the best case and 10^2–10^{13} elsewhere. Setting suppression aside, the required $c = 0.145$–0.751 means $\Gamma = cE$, a process near the unitarity limit, against $\Gamma \sim 10^{-3}E$ for a perturbative gauge interaction. Within the local thermal Hagedorn-production ansatz tested here, the required conversion rate is incompatible with EFT-controlled sub-string-scale transition densities — the same pattern, and the same cause, as the earlier closure of the route to ξ: the string scale sits orders of magnitude above the bounce scale. The scope is stated explicitly: gravitational particle production, non-thermal production such as parametric resonance, tachyonic instabilities at enhanced-symmetry points, winding-mode annihilation and non-perturbative effects were none of them computed, so this excludes no string-theoretic mechanism in general.

7.12 Deriving the entropic mass: a partial success

Asking where the tuned λ comes from gives a partial answer. First, λ is not free: demanding exact scale invariance gives the closed form $\lambda_{\text{SI}} = 2(p_i - 1)^2 + p_i - 2$, matching the numeric solver to machine precision (exactly 104 at $p_i = 8$), so choosing p_i fixes it. Second, the *form* is derived: in ekpyrotic contraction $R = 6(2 - p)H^2 < 0$ for $p > 2$, so a standard non-minimal coupling $\xi R s^2$ gives $m^2_{\text{eff}} = 6\xi(2 - p)H^2$, automatically tachyonic and automatically proportional to H^2 — the ansatz did not need inserting by hand. Third, the *value* is not, because the scalings disagree: every natural source is linear in p ($|R|/H^2 = 6(p - 2)$, $\dot{\sigma}^2/H^2 = 2p$, $|V|/H^2 = p - 3$) while the requirement $\simeq 2p^2$ is quadratic, so the needed coefficient grows linearly and at $p_i = 8$ reaches $\xi = 26/9 = 2.889$, seventeen times the conformal value. Lowering p makes ξ ordinary but drives the starting density down eight orders of magnitude, and raising it does the reverse; the 0.29% tuning transfers intact to whatever coefficient supplies it. Incidentally, at the optimum $\rho_{\text{tr}} = 0.3\rho_c$ the anisotropy budget needs only $\rho_i = 1.2 \times 10^{-6}\rho_c$ rather than $2 \times 10^{-10}\rho_c$ — four orders of magnitude of relief. The route did not close as it did for the conversion rate, but a tuned $O(1)$ number remains, so the claim that this framework is simpler than inflation does not hold.

7.13 Relation to prior work

None of this establishes a new law. That vacuum-initial-condition ekpyrotic contraction yields a strongly blue adiabatic spectrum is an established property of the scenario, not a finding of this work: Lyth showed the collapsing-phase spectrum to be strongly scale-dependent [15], and the point is standard in the subsequent literature [16, 12]. What is added is the quantification that the derived background makes it worse. That stability and effective-field-theory control are the central difficulties for non-singular bounces is likewise the subject of an existing literature [13, 14]; the latter is directly relevant here, since it treats exactly the loop-quantum-cosmology-like effective Friedmann equation used in this paper and asks when an effective field description reproduces it. The coefficient gate above is an internal consistency check, and no covariant action generating the specific $w(t)$ has been derived. That a Hagedorn phase does not by itself solve flatness is consistent with existing string-gas discussions. What is added is the combination of $\rho_s/\rho_c > 725$ with $\rho_{\text{tr}} \lesssim 0.3\rho_c$ to exclude the thermal Hagedorn route quantitatively, and nothing beyond that scope is claimed.

7.14 A candidate covariant action, and two obstructions

This is the item named as most needed. The natural candidate is standard coupled quintessence: an ekpyrotic scalar with a pressureless sector coupled to it conformally, $A(\phi) = e^{\beta\phi}$ in $S_m[\psi, A^2(\phi)g_{\mu\nu}]$, giving $\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = -\beta\rho_2$ and $\dot{\rho}_2 + 3H\rho_2 = +\beta\dot{\phi}\rho_2$, whose sum conserves the total energy identically.

A covariant origin for the transfer rate does exist: $\Gamma_{\text{eff}} = \beta\dot{\phi}$, and on the attractor $\dot{\phi}^2 = 2p\rho/3$, so $\Gamma_{\text{eff}} \propto H$. The Hubble scaling used above is what a conformal coupling supplies for free, and the earlier characterisation of it as a gravitational ansatz without local justification went too far: that exclusion applies only to a *thermal* origin for the rate. The density threshold also disappears, the crossover occurring by itself whenever $\beta < (c/2)(3/p - 2) = -3.25$.

Two obstructions then block each other. With an unbounded potential the pressureless sector can hold 94% of the energy (at $\beta = -8$) while $p_\phi = K - V \geq -V$ diverges alongside it, leaving $w_{\text{eff}} = 9.16$: draining $\rho_\phi = K + V$ does not drain p_ϕ. Bounding the potential below caps p_ϕ but makes the field pass the minimum and turn around, so $e^{\beta\phi}$ stops growing — all fifteen combinations tested land on $w = 1.000$, kination. Since $p_\phi \geq -V$, a falling w requires V bounded below, and V bounded below switches off the coupling that was to drive the transfer.

Something more important surfaces here: the two-fluid parametrisation used above is *not faithful to a scalar realisation.* Writing $p = w\rho$ per component is a definition for a fluid but does not hold for a field, and that difference decides whether the transition happens. The results obtained from the two-fluid model must therefore be re-read as conditional on such a fluid existing. What was tested is a canonical scalar with a conformal coupling and two families of exponential potential; derivative or disformal couplings, non-canonical kinetic terms, more than two fields, and the possibility that the beyond-Horndeski sector itself carries the transition were not tested. The last is the most promising, since the coefficient gate has already confirmed that such a completion exists.

7.15 The beyond-Horndeski sector does not carry the transition

This was the most promising route left above, and it does not hold up. The Ye–Piao construction is a *reconstruction*: it takes $H(t)$ and $a(t)$ as input and solves algebraically for the Lagrangian functions, without predicting $H(t)$. That an action exists for the derived background therefore means only that one exists for any background fed to the algebra. For the sector to be informative, the stability gate would have to be selective.

The comparison must be controlled. Swapping $p(\eta)$ inside the closed-form ansatz $H = \eta/[p(1+\eta^2)]$ also fixes $\rho/\rho_c = 1/(1+\eta^2)$ and so changes $H(t)$: fitting a tanh to the derived background's own $p(t)$ to 1.2% and feeding it through that ansatz drops the viable count from 4,611 to zero. The comparison backgrounds were therefore built by integrating the same equations with only $w(\rho)$ replaced, on a grid wide enough to contain every transition.

background	viable	fraction	claimed
derived	4611	6.19%	yes
reversed transition $w : 0 \to 4.33$	158	0.21%	no
constant $w = 4.33$	0	0.00%	no
constant $w = 0$	5970	8.01%	no

The gate is not vacuous — constant $w = 4.33$ fails everywhere tested — but neither is it selective: a background with *no transition at all* passes more easily than the derived one. Passing it is therefore not evidence that the background is right, and this sector cannot be the origin of $w(t)$. What discriminates is anisotropy: constant $w = 0$ has $1 - 3/p = -1$ and grows shear throughout, so it is excluded by the shear requirement, not by stability. This route closes for a different reason than ξ or γ did — there a mismatch of scales blocked the way, here the obstruction is methodological, since a reconstruction returns what was put in.

7.16 A derivative coupling: a promising candidate, not a covariant completion

Obstruction 2 above was that bounding the potential makes the field turn around, after which $e^{\beta\phi}$ stops growing. Coupling to the *kinetic term* rather than the field value removes it, since the kinetic energy keeps growing through contraction whether or not ϕ turns around. Take

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{2} - \frac{X}{2} - V(\phi)\right] + S_m[\psi, A^2(X)g_{\mu\nu}], \qquad \ln A = \frac{\nu}{2}X, \quad X = \dot{\phi}^2, \tag{18}$$

for which total energy conservation gives

$$\dot{\rho}_2 + 3H\rho_2 = \nu\dot{\phi}\ddot{\phi}\,\rho_2, \qquad \ddot{\phi}\,(1+\nu\rho_2) = -3H\dot{\phi} - V_{,\phi}. \tag{19}$$

The driver $\dot{\phi}\ddot{\phi}$ is positive whenever the kinetic energy grows, so the obstruction is evaded by construction, and the X-dependence appears as an effective kinetic normalisation $(1+\nu\rho_2) > 0$. The bounded potential required by obstruction 1 is now permitted.

The descent happens: raising ν from 100 to 10^4 drives the final p monotonically from 3.000 to 1.504, with the pressureless sector dominating (f_{ek} down to 0.25%). Three costs attach. Anisotropy trades against the depth of the descent — at $\nu = 10^4$ the budget reaches +0.401 and suppression is gone, the compromise being $|V_{\min}| = 10^{-2}$, $\nu = 10^3$, giving $p = 1.541$ and a budget of -3.360. At the bounce $1 + \nu\rho_2 \sim 10^3$, so the coupling is not perturbative. And the required $\ln A \simeq 27$ is set by the initial pressureless abundance: a 10% initial fraction needs only $\ln A = 10.4$, at the price of a budget of -2.291.

The background equations above are *effective equations inspired by* a candidate action, not the result of varying it. Symbolic computation established two things: given Q, total energy conservation fixes the ϕ equation uniquely and reproduces the form used, so the system is internally consistent; but the coupling term one would expect from varying the matter action changes the denominator from $(1 + \nu\rho_2)$ to $(1 + \nu\rho_2 + \nu^2\rho_2 X)$, and near the bounce $\nu X \simeq 55$, so the two differ by about fifty-five fold. Since $X = -g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ already depends on the metric and on $\partial\phi$, varying $S_m[\psi, A^2(X)g_{\mu\nu}]$ produces terms beyond a simple two-fluid transfer that were not pinned down.

The stability gate is therefore withheld: with the background unsettled at that magnitude, computed values of Q_s, c_s^2, Q_T, c_T^2 would not say what they belong to. Four verifications are required — re-deriving the exact background equations by variation; checking the DHOST degeneracy conditions against an Ostrogradsky ghost; obtaining the quadratic action on that background; and confirming that $1 + \nu\rho_2 \sim 10^3$ does not break strong coupling or the EFT cutoff. What can be said is that a bounded potential with a kinetic-dependent coupling permits a $p_f \simeq 3/2$ transition, that it evades the obstruction of the ϕ-conformal coupling, and that this is the first numerical indication that the fluid model may carry over to a field. What cannot be said is that a covariant action has been found; the result is recorded as a candidate realisation, a first viable derivative-coupling proxy.

7.17 What this section does not claim

1. The effective-field-theory cutoff is not derived.
2. No microscopic mechanism fixes the relic amplitude; only its shape is predicted.
3. The entropy field and its 0.29% tuning are inserted by hand.
4. Section 7.7 narrows the *class* of dynamics that can drive the transition but does not fix the conversion rate: $\gamma \simeq 26$ and ρ_H remain inputs rather than quantities computed from a string-excitation production rate.
5. The two derivation attempts end in a failure and a partial success: string production is excluded for the conversion rate, and for the entropic mass only the form is derived, the value $\xi = 2.889$ remaining a tuned $O(1)$ number. The threshold $\rho_{\rm tr}$ is still an input.
6. No data analysis was carried out. The analysis design is pre-registered, but no CMB map was opened.

8 Limitations and next steps

- **Open.** A microscopic derivation of η and ξ. The EFT cutoff Λ belongs here: it is the one entry of the stability gate that remains an assumption.
- **Partly settled.** The dynamics of the equation-of-state transition. Constant-w matter and a scalar oscillating about a minimum are both excluded, leaving energy conversion above a density threshold; that route predicts $p_f = 3/2$ and squeezes the threshold from both sides, but does not fix the rate.
- **Open.** The microscopic origin of the conversion rate. Local thermal Hagedorn production is excluded; what remains is the requirement of a process near the unitarity limit at 10^{16}–10^{17} GeV, for which no candidate exists.

- **Partly settled.** The entropic mass. Its *form* follows from a non-minimal coupling $\xi R s^2$; its *value* $\xi = 2.889$ has no independent justification, must be tuned to 0.29%, and must be re-solved whenever the background changes.

- **Open.** A covariant action generating the specific $w(t)$. The coefficient gate is an internal consistency check rather than a derivation, and it passes a background with no transition more easily than the derived one. Of three routes tested, a conformal coupling and the beyond-Horndeski sector both close; only a derivative coupling survives as a candidate, with four checks outstanding — varying the full action, the DHOST degeneracy conditions, the quadratic action of perturbations, and strong coupling at $1 + \nu\rho_2 \sim 10^3$.

- **Open.** The reliance of the transition results on a *fluid* description. Writing $p = w\rho$ per component defines a fluid but is not a property of a field, and that difference decides whether the transition occurs; a first indication that it can also happen for a field exists but is not settled.

- **Partly settled.** Stability including anisotropies. Scalar and tensor second-order perturbations are decided over the whole range; anisotropy is computed only at the level of a perturbative test shear.

- **Open.** Global matching from a black-hole interior to an expanding region.

- **Open.** Information, entropy and Hawking radiation.

- **Partly settled.** Observables comparable with data. The scalar tilt and the quadrupole shape are computed; the amplitude is not, and the gravitational-wave side is absent. On the derived background the adiabatic tilt *worsens*, from $n_s = 3.34$ to 4.54.

Conclusion

A finite curvature or density scale remains a useful research hypothesis for replacing a classical singularity with a non-singular transition. The verification reported here moved the proposal in two directions at once. It advanced, in that the effective background belongs to a family with a stable microscopic realisation, and that a primordial spectrum and a quadrupole shape follow from it by calculation. It narrowed, in that the same calculation excluded the minimal single-field realisation, the constant-equation-of-state baseline, and the standing of the bounce's adiabatic mode as the origin of the CMB.

What survives is a single quadrupolar signal whose shape is predicted and whose size is not, resting on an anisotropy put in by hand.

Replacing the remaining hand-inserted entries with calculations gained three things and lost three. Gained: two exact arguments excluding constant-w matter and an oscillating scalar from driving the transition, with the surviving route predicting $p_f = 3/2$; a coefficient gate that passes again on that background; and the *form* of the entropic-mass ansatz, which a non-minimal coupling supplies for free. Lost: an adiabatic tilt that gets worse rather than better, $n_s = 3.34 \to 4.54$; the route to the conversion rate through local thermal Hagedorn production, closed because EFT control pushed the bounce below the string scale while the transition must sit further below still; and the transferability of both the coefficients and the tuned value, neither of which carries over when the background changes.

What remains unknown is no longer a set of free functions but three numbers — the transition threshold, the conversion rate, and the coupling ξ. That is progress, but while a tuned $O(1)$ number remains, the claim that this framework is simpler than inflation does not hold. A condition prior to all three also surfaced: the two-fluid parametrisation from which these results were obtained is not faithful to a scalar realisation, so they are conditional on such a fluid existing. The value of this work lies not in declaring an answer to quantum gravity, but in recording which of the required items turned out to be computable — and which did not survive being computed. The single thing most needed is still a covariant action generating the specific $w(t)$. The most natural candidate fails as described above, and the beyond-Horndeski route closes for a methodological reason: a reconstruction returns what was put in, and its gate passes a background with no transition more easily than the derived one. From that list a derivative coupling remains a promising candidate, as described below: coupling to the kinetic term keeps the transfer alive after the field turns around, resolving both obstructions and driving the final p to $3/2$. Its background equations have not, however, been obtained by varying an action, and they differ from the expected form by about fifty-five fold near the bounce, so a covariant action has not been found and the stability gate is withheld.

Data and code availability

All numerical results were produced by scripts that read no observational data. Analysis code, gate criteria, simulation outputs and the pre-registered CMB analysis design are maintained with the manuscript and are available from the author on request.

References

[1] G. Ye and Y.-S. Piao, "Implication of GW170817 for cosmological bounces," *Commun. Theor. Phys.* **71**, 427 (2019), arXiv:1901.02202.

[2] T. Kobayashi, "Generic instabilities of non-singular cosmologies in Horndeski theory: a no-go theorem," *Phys. Rev. D* **94**, 043511 (2016), arXiv:1606.05831.

[3] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.

[4] J. Kim and E. Komatsu, "Limits on anisotropic inflation from the Planck data," *Phys. Rev. D* **88**, 101301 (2013), arXiv:1310.1605.

[5] Planck Collaboration, "Planck 2018 results. VII. Isotropy and statistics of the CMB," *Astron. Astrophys.* **641**, A7 (2020), arXiv:1906.02552.

[6] J. Khoury, B. A. Ovrut, P. J. Steinhardt and N. Turok, "The ekpyrotic universe: colliding branes and the origin of the hot big bang," *Phys. Rev. D* **64**, 123522 (2001), arXiv:hep-th/0103239.

[7] R. H. Brandenberger, "String gas cosmology: progress and problems," *Class. Quantum Grav.* **28**, 204005 (2011), arXiv:1105.3247.

[8] D. Battefeld and P. Peter, "A critical review of classical bouncing cosmologies," *Phys. Rep.* **571**, 1 (2015), arXiv:1406.2790.

[9] M. Novello and S. E. P. Bergliaffa, "Bouncing cosmologies," *Phys. Rep.* **463**, 127 (2008), arXiv:0802.1634.

[10] R. Penrose, "On the gravitization of quantum mechanics 2: conformal cyclic cosmology," *Found. Phys.* **44**, 873 (2014).

[11] L. McAllister and E. Silverstein, "String cosmology: a review," *Gen. Relativ. Gravit.* **40**, 565 (2008), arXiv:0710.2951.

[12] Y.-F. Cai, D. A. Easson and R. Brandenberger, *Towards a Nonsingular Bouncing Cosmology*, JCAP **08** (2012) 020, arXiv:1206.2382.

[13] A. Ijjas, *Space-time slicing in Horndeski theories and its implications for non-singular bouncing solutions*, JCAP **02** (2018) 007, arXiv:1710.05990.

[14] Y.-F. Cai and E. Wilson-Ewing, *Non-singular bounce scenarios in loop quantum cosmology and the effective field description*, JCAP **03** (2014) 026, arXiv:1402.3009.

[15] D. H. Lyth, *The primordial curvature perturbation in the ekpyrotic Universe*, Phys. Lett. B **524** (2002) 1, arXiv:hep-ph/0106153.

[16] J.-L. Lehners, *Ekpyrotic and Cyclic Cosmology*, Phys. Rept. **465** (2008) 223, arXiv:0806.1245.